

دانا ابراهيم السكيّتي

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Part 1: Capturing HTTP Traffic.

```
[Next Sequence Number: 246      (relative sequence number)]
      Acknowledgment Number: 1      (relative ack number)
      Acknowledgment number (raw): 1841609304
      Header Length: 20 bytes (5) = .... 0101
      Flags: 0x018 (PSH, ACK) <
      Window: 255
      [Calculated window size: 255]
[Window size scaling factor: -1 (unknown)]
      Checksum: 0x52c6 [unverified]
      [Checksum Status: Unverified]
      Urgent Pointer: 0
      [Timestamps] <
      [Client Contiguous Streams: 0]
      [Server Contiguous Streams: 1]
      TCP payload (245 bytes)
      Hypertext Transfer Protocol
GET /msdownload/update/v3/static/trustedr/en/disallowedcertstl.cab?83c735c032dbce8e HTTP/1.1\r\n <
      Cache-Control: no-cache\r\n
      Connection: Keep-Alive\r\n
      Pragma: no-cache\r\n
      Accept: */*\r\n
      User-Agent: Microsoft-CryptoAPI/10.0\r\n
      Host: ctldl.windowsupdate.com\r\n
      r\n\r\n
Full request URI: http://ctldl.windowsupdate.com/msdownload/update/v3/static/trustedr/en/disallowedcertstl.cab?83c735c032dbce8e]
```

Part 2: Analyzing TCP/IP Traffic.

	Info	length	Protocol	Destination	Source	Time	No.
Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM [SYN] 80 → 55372	66	TCP	199.232.210.172	10.29.131.189	32.272303700	628	
Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 SACK_PERM WS=512 [SYN, ACK] 55372 → 80	66	TCP	10.29.131.189	199.232.210.172	32.303626800	631	
Seq=1 Ack=1 Win=65280 Len=0 [ACK] 80 → 55372	54	TCP	199.232.210.172	10.29.131.189	32.303714200	632	
GET /msdownload/update/v3/static/trustedr/en/disallowedcertstl.cab?8723874264a81af8 HTTP/1.1	299	HTTP	199.232.210.172	10.29.131.189	32.303800300	633	
55372 → 80 [PSH, ACK] Seq=1 Ack=1 Win=65280 Len=245 [TCP Retransmission]	299	TCP	199.232.210.172	10.29.131.189	32.559195200	639	
55372 → 80 [PSH, ACK] Seq=1 Ack=1 Win=65280 Len=245 [TCP Retransmission]	299	TCP	199.232.210.172	10.29.131.189	32.860743900	651	
55372 → 80 [PSH, ACK] Seq=1 Ack=1 Win=65280 Len=245 [TCP Retransmission]	299	TCP	199.232.210.172	10.29.131.189	33.474098200	728	
55372 → 80 [PSH, ACK] Seq=1 Ack=1 Win=65280 Len=245 [TCP Retransmission]	299	TCP	199.232.210.172	10.29.131.189	34.678340400	876	
55372 → 80 [PSH, ACK] Seq=1 Ack=1 Win=65280 Len=245 [TCP Retransmission]	299	TCP	199.232.210.172	10.29.131.189	37.080028500	922	
55372 → 80 [PSH, ACK] Seq=1 Ack=1 Win=65280 Len=245 [TCP Retransmission]	299	TCP	199.232.210.172	10.29.131.189	41.883679700	997	
Seq=246 Ack=1 Win=0 Len=0 [RST, ACK] 80 → 55372	54	TCP	199.232.210.172	10.29.131.189	51.488873300	1216	

Part 3: Capturing and Analyzing UDP Traffic.

UDP header is very simple compared to TCP header.

User Datagram Protocol, Src Port: 60313, Dst Port: 53

Source Port: 60313

Destination Port: 53

Length: 49

Checksum: 0x66d3 [unverified]

[Checksum Status: Unverified]

[Stream index: 0]

[Stream Packet Number: 3]

[Timestamps]

UDP payload (41 bytes)

Part 4: Comparing TCP and UDP.

Task 1: Fill in the following table and provide reasons.

	TCP or UDP	Reasons
Reliability and Connection Establishment	TCP	guarantees packet delivery via acknowledgments and retransmissions.
Data Integrity and Ordering	TCP	Uses sequence numbers and checksums/ acknowledgments to ensure data integrity.

Task 2: Identify the use Cases and Performance of TCP and UDP.

	TCP	UDP
Use cases	Web browsing (HTTP), File transfer (FTP), Emails.	Live streaming, Online gaming, VoIP, DNS.
Performance	Slower due to handshakes and error checks.	Faster with lower latency and no setup delay.

